

Discussion #13 *Solutions*

Learning Objectives

Principal Component Analysis (PCA)

- Distinguish supervised vs. unsupervised learning; explain why dimensionality reduction is unsupervised
- Define intrinsic dimension, column-space dimension, and rank of a matrix
- Distinguish principal components and latent features/factors.
- Interpret PCA plots, scree plots, and biplots

Clustering

- Describe the k-means algorithm; define inertia and distortion; define silhouette score
- Define hierarchical agglomerative clustering; distinguish single, average, and complete linkage
- Describe how to choose K using the elbow method, dendrograms, and silhouette scores; interpret a dendrogram and silhouette plot

Disclaimer: Think of these learning objectives as a high-level guide to help you organize your studying. This is not a comprehensive list of every topic covered in lecture or discussion.

PCA Basics

1. James is conducting a research study and collects the below data for further analysis. Consider the following dataset, where X is the corresponding 4×3 design matrix.

Observations	Feature 1	Feature 2	Feature 3
1	7.39	-3.59	-0.78
2	-5.32	-8.37	0.90
3	-0.61	1.75	-0.62
4	-1.46	10.21	0.50

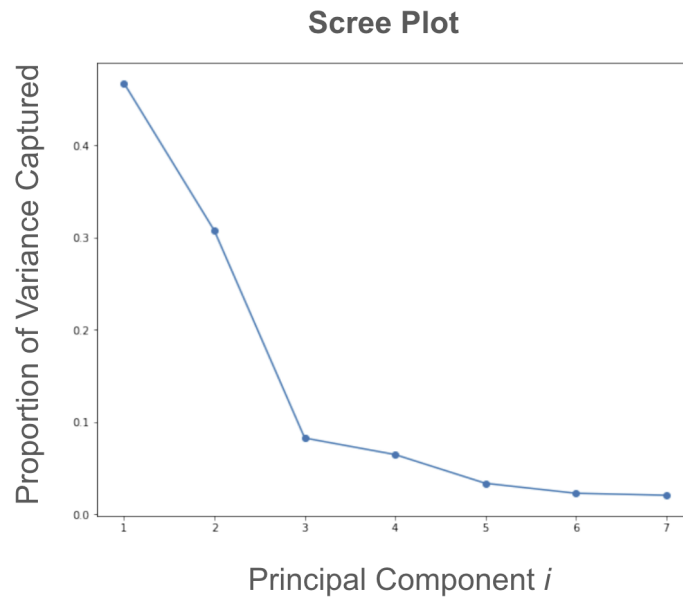
Suppose we obtain the following matrix of principal components.

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- (a) Compute the projection of X onto the first principal component, also called the first latent factor (round to 2 decimal places).

Solution: We compute the first latent factor by multiplying X by the first column of the principal component matrix (PC1) to get $[-3.59 \quad -8.37 \quad 1.75 \quad 10.21]^T$.

James realizes that his dataset is too small! He decides to collect additional data on 50 observations, this time selecting 7 features instead of 3. The scree plot below depicts the proportion of variance captured by each principal component for this new dataset.



- (b) Using the elbow method, how many principal components should we choose to represent the data?

Solution: The elbow of a scree plot is the point where adding more principal components results in diminishing returns. In other words, we look for the “elbow” in the curve just before the line flattens out. This point is at PC3 in the graph above, so we choose 3 as the number of principal components. Note that not all scree plots will have an obvious elbow.

- (c) Evaluate the statements below based on the scree plot obtained by James.

TRUE False Latent features are projections of original data onto these principal components.

True **FALSE** The rank of this dataset is less than 3.

True **FALSE** There can be 8 principal components for this new dataset.

Solution: The first statement is true. Principal components are themselves the “recipe” for constructing latent factors from the original data. In other words, the latent factors constructed by PCA are linear combinations of the original features. The second statement is false. A dataset’s rank is at least as much as the number of principal components that capture nonzero proportion of variance. Even beyond the “elbow” of the plot, additional principal components capture nonzero variance. The third statement is false. There can be at most 7 principal components for cooks because it has 7 columns. The number of principal components a dataset can have is upper-bounded by its number of columns.

Note: The scope of PCA this semester differs substantially from semesters prior to Fall 2025. This semester, you are not expected to know how to complete any questions related to singular value decomposition, matrix factorization, or reconstruction error.

Interpreting PCA Plots

2. Dan has three datasets A , B , and $C \in \mathbb{R}^{100 \times 2}$. That is, each dataset consists of 100 data points in two dimensions. He visualizes the datasets using scatterplots, labeled Plot A, Plot B, and Plot C, respectively:



- (a) For which dataset does the second principal component capture the least variance?

- ☐ A. Dataset A
☒ B. Dataset B
☐ C. Dataset C
☐ D. Cannot determine with the given information

Solution: Most of the variance in Plot B lies roughly about a single line. The variance captured by the second PC is far smaller in Plot B compared to plots A and C. In both Plot A and Plot C, there are significant amounts of variance in two orthogonal directions.

- (b) For one of the three datasets, Dan obtains a PC2 vector of $[a, b]^T$, where $a > 0$ and $b > 0$. Which of the following datasets could reasonably match this principal component vector?

- ☐ A. Dataset A
☒ B. Dataset B
☐ C. Dataset C
☐ D. None of the above

Solution: The second principal component is the direction of second-largest variance in each plot. PC2 should also be orthogonal to PC1. Only in plot B does the direction of PC2 align with the positive slope diagonal, making $a > 0$ and $b > 0$ feasible constraints.

- (c) Dan discovers that in plot C, the direction of PC1 is best approximated by $[1, 1]^T$. List some examples of vectors that the direction of PC2 can reasonably be approximated by before normalization.

Solution: Any vector that is orthogonal to $[1, 1]^T$ (i.e. dot product is zero) is acceptable!

- (d) Dan consults with other members of course staff, who inform him that the variance of each feature in Dataset B is 20. What is the sum of the variance of Dataset B in the PC1 dimension and the variance of Dataset B in the PC2 dimension?

Solution: The number of principal components, two, matches the number of original features in each dataset. Thus, the sum of the variance along each principal component should match the total variance of the original dataset B, which would equal $20 \times 2 = 40$.

What is Clustering?

3. (a) Describe the difference between clustering and classification.

Solution: Both involve assigning arbitrary classes to each observation. In classification, we have a labeled training set, i.e. a ground truth we can use to train our model. In clustering, we have no ground truth, and are instead looking for patterns in unlabeled data. Classification falls under supervised machine learning because it is concerned with predicting the correct/specific class given the true labels to the data. Clustering falls under unsupervised machine learning because there are no provided “true” labels - instead it is concerned with the patterns across the data (rather than where the data belongs).

- (b) Given a set of points and their labels (or cluster assignments) from a K-Means clustering, how can we compute the centroids of each of the clusters?

Solution: For each label, find the mean of all of the data points corresponding to that label. In other words, compute centroids c_i for each label/cluster's set of data points L_i and corresponding data points $x_j \in L_i$:

$$c_i = \frac{1}{n} \sum_{x_j \in L_i} x_j$$

- (c) Suppose that no two points have the same distance from each other. Are the cluster labels computed by K-means always the same for a given dataset? What about for max agglomerative clustering?

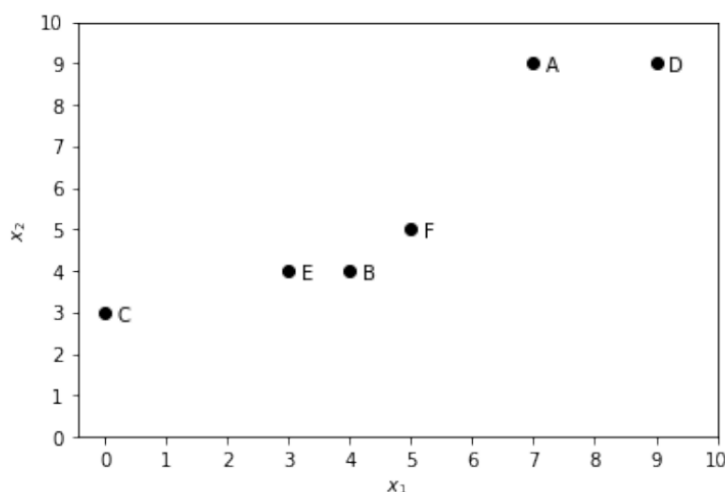
Solution: The cluster labels computed by K-means could be different if we change the initial centers. For example, given 3 1-D points $\{0, 4, 10\}$ and K-means with $k = 2$. If the initial centers are 0 and 10, the algorithm converges to clusters $\{0, 4\}$, and $\{10\}$. Whereas if the initial centers are 0 and 4, the algorithm converges to clusters $\{0\}$ and $\{4, 10\}$. Thus different initializations can give different final cluster labels.

This cannot happen for max agglomerative clustering because every point starts in its own cluster, and there is no choice involving initialization.

Clustering Mechanics

4. Rohit has the following dataset, where each point has two features, x_1 and x_2 .

x_1	x_2	Point Label
7	9	A
4	4	B
0	3	C
9	9	D
3	4	E
5	5	F



(a) Rohit decides to perform k-means clustering on this dataset with 2 clusters: the left cluster has an initial center at (3, 3), and the right cluster has an initial center at (10, 10).

- i. Which points belong to the left cluster after the first iteration? Format your answer as a list of point labels separated by commas in alphabetical order.

Solution: B, C, E, F. These points are closer to (3, 3) than (10, 10).

- ii. Where are the clusters centered after the first iteration?

Left Cluster Center = (,) Right Cluster Center = (,)

Solution: (3, 4) and (8, 9). These are the average x_1 and x_2 values for the respective clusters.

- iii. Which points belong to the left cluster after the second iteration? Format your answer as a list of point labels separated by commas in alphabetical order.

Solution: B, C, E, F. These points are still closer to the new left center than the new right center.

- iv. How many points do we expect to change clusters if Rohit were to perform a third iteration?

Solution: 0. The points in the two clusters remained the same between the first and second iterations, meaning that k-means converged.

- (b) Rohit now wants to try hierarchical agglomerative clustering with **single linkage**. After the **third** iteration, how many clusters exist, and which points are in each cluster?

Solution: In hierarchical agglomerative clustering, each data point starts off in its own cluster, and we iteratively merge "clusters" together according to a specific linkage criterion. In this case, we use single linkage, so the **minimum distance** between two points in different clusters determines the next merge.

At iteration 1, we merge E and B together. At iteration 2, we merge F with the newly created [B, E] cluster.

At iteration 3, it becomes a little trickier to visually determine the closest clusters. Our top contenders are merging C with [B, E, F], merging A with [B, E, F], or merging A with D. Using single linkage criteria, the distance between C and [B, E, F] reduces to the distance between C and the closest point in [B, E, F], which is E. The distance between A and [B, E, F] reduces to the distance between A and the closest point in [B, E, F], which is F. The distance between C and E is $\sqrt{(3-0)^2 + (4-3)^2} = \sqrt{10}$. The distance between A and F is $\sqrt{(7-5)^2 + (9-5)^2} = \sqrt{20}$. Finally, the distance between A and D is $\sqrt{(9-7)^2 + (9-9)^2} = \sqrt{4}$. Thus, we merge A and D together.

We ultimately end our third iteration with 3 clusters: [B, E, F], C, and [A, D].

- (c) Suppose now, Rohit is working with the same dataset, but with new clusters. One cluster has points [B, C, E], and the other cluster has points [A, D, F].

- i. What is the distance between these two clusters by **complete linkage**?

Solution: We take the distance between the two farthest data points, which would be C and D. This distance would be $\sqrt{(9-0)^2 + (9-3)^2} = \sqrt{117}$.

- ii. Rohit also wants to understand how well point F fits into its own cluster. Using the formula for **silhouette score**, $S = \frac{B-A}{\max(A, B)}$, where A is the mean distance to other points in its cluster, and B is the mean distance to points in its closest cluster, she calculates a score for point F of -0.41 . Interpret this result.

Solution: Silhouette scores help us assess how well a point fits into its cluster. A higher score indicates that B is far higher than A , so it is on average much closer to points in its own cluster than points in the nearest cluster. A negative silhouette score indicates that point F is on average closer to points in the nearest cluster than its own clustermates! This point is likely misclassified.

- (d) Which of the following statements are true regarding clustering? Select all that apply.

☐ A. If we run agglomerative clustering with the same k and the same type of linkage, we will still get the same results each time.

- ☐ B. Distortion is a version of inertia weighted by cluster size.
- ☐ C. When choosing the optimal k for k-means clustering, we should pick the k with the lowest L_2 loss.
- ☐ D. Within one iteration of k-means clustering, some clusters' centers may move even if other centers do not.

Solution:

Option A is correct; k-means clustering is the type of clustering that can vary depending on the initial centers.

Option B is the correct definition of distortion.

Option C is incorrect. We should use the elbow method to select the optimal k .

Option D is correct. Imagine you have 3 clusters, one very far from the other two. The other two may swap points between one another and change their centers, while the far cluster remains the same.